

Difference Engine in 1822 were to play key roles in the future of automated robots. Another significant development was the patenting of a remote controlled robot boat in 1898 by Nikola Tesla, which would form the basis of the wireless control that we see in space exploration robots even today.

In the 1921 play R.U.R (Rossum's Universal Robots), writer Karel Capek used the word 'Robot' to denote automatons in the story. He credits the word to his brother, who derived it from the Czech word 'robota' that means servitude (kind of setting the intent right there). But the word 'robotics' is said to have come into common parlance due to someone else. Asimov's science fiction writing in the 1940s were a major, and perhaps the most significant, cultural phenomenon when it comes robotics. Most importantly, he conceived the three laws of robotics:

- **First Law:** A robot may not injure a human being, or, through inaction, allow a human being to come to harm.
 - **Second Law:** A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
 - **Third Law:** A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.
- He also added a zero law later:
- **Zeroth Law:** A robot may not harm humanity, or, by inaction, allow humanity to come to harm.

In 1928, Japan's first humanoid robot was designed and built by Makoto Nishimura, a biologist and it was called the Gakuten-soku. This was again a mecha-noid meant for display purposes as it could change its expression and perform some gestures such as moving its head and hand. Another significant humanoid built by the Westinghouse Electric Corporation somewhere between 1937 and 1938 was Elektro, a robot. This one could blow balloons, speak about 700 words (using a record player), walk by voice command, smoke cigarettes, and



Elektro with Sparko